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Machine Learning-based Approach for Scam Message Detection in Kiswahili Using Natural Language

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Scam messages transmitted via SMS and digital platforms present significant cybersecurity threats, particularly in regions where mobile money services are widely used. In Tanzania, where Kiswahili is the dominant language of communication, existing scam detection systems—primarily designed for English—fail to capture the linguistic and contextual nuances of Kiswahili messages. This study presents a machine learning-based approach for detecting scam messages written in Kiswahili using Natural Language Processing (NLP) techniques. A labelled dataset of 5,000 real-world messages was compiled through public sources, user contributions, and collaboration with law enforcement. Four models—Support Vector Machine (SVM), Random Forest (RF), Bidirectional Long Short-Term Memory (Bi-LSTM), and Swahili-BERT—were developed and evaluated using accuracy, precision, recall, and F1-score. Swahili-BERT outperformed all other models, achieving a detection accuracy of 96.3%, demonstrating the efficacy of transformer-based models in low-resource language settings. The system was deployed through a Flask-based API for real-time use on mobile platforms. This research advances Kiswahili NLP, strengthens digital security infrastructure in Tanzania, and offers a scalable solution to combat SMS-based fraud in mobile money ecosystems.

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INTRODUCTION

Cybersecurity has become an essential pillar in protecting digital communication systems from evolving threats such as scam messages, phishing attacks, and data breaches. Scam messages designed to deceive individuals into revealing sensitive information pose significant risks to personal privacy and financial security. Traditional detection approaches, such as rule-based systems, face limitations in scalability and adaptability [1]. Advances in NLP and ML offer promising alternatives by enabling automated detection of scam content through linguistic and contextual analysis. In Tanzania, mobile money platforms dominate financial transactions, and Kiswahili is the most widely used language in digital communication. This creates a unique context where scam detection systems tailored to Kiswahili are urgently needed. While English-language scam detection research is advanced, Kiswahili remains underrepresented as a low-resource language in NLP.

Despite the increasing prevalence of scam messages in Tanzania, there remains a lack of automated detection systems specifically designed for Kiswahili [2]. Most existing models are optimised for English and fail to account for the linguistic nuances of Kiswahili scam messages, leaving millions of mobile money users vulnerable to fraud [2]. The absence of annotated datasets and specialised NLP tools for Kiswahili further constrains system development.

This study set out to develop NLP-driven machine learning models capable of detecting scam messages written in Kiswahili. To achieve this, the research investigated linguistic patterns within scam texts, constructed a labelled dataset, and trained multiple models: Support Vector Machines (SVM), Random Forest, Bi-LSTM, and Swahili-BERT, which were evaluated using

standard performance metrics. The study also examined the perceptions and expectations of key stakeholders in Tanzania, particularly the Tanzania Police Force. Therefore, the study focuses on scam messages written in Kiswahili, transmitted via SMS and mobile platforms. It excludes voice-based or image-based scams.

The remainder of this paper is structured as follows: The background information of this study and the related works will be stated in Section 2; the proposed ML Approach will be stated in Section 3; and the research methodology will be presented in Section 4. The results will be presented in Section 5, the discussion will be described in Section 6, and finally, Section 7 will conclude this study and provide recommendations for future directions.

BACKGROUND INFORMATION AND RELATED WORKS

Scam detection through NLP and ML has been extensively studied for English and other high-resource languages. However, Kiswahili presents unique challenges due to a lack of annotated datasets, informal language usage, and morphological complexity. The intersection of Natural Language Processing (NLP) and machine learning (ML) in scam detection has seen substantial progress in recent years, particularly in high-resource languages such as English and Chinese. However, the application of such technologies to low-resource languages like Kiswahili remains limited. This section reviews prior research relevant to three major themes of this study: scam detection via ML, Kiswahili language processing, and cybersecurity in mobile financial platforms.

Significance - This research study significantly enhances cybersecurity for Swahili speakers, informs policy and regulation, empowers telecom

and financial service providers, advances NLP research in low-resource languages, and contributes practical tools for detecting fraudulent messages.

RELATED WORKS

Scam Detection via NLP and Machine Learning

Several studies have explored the use of ML classifiers and NLP techniques to detect fraudulent content in text messages, emails, and websites. Traditional classifiers such as Support Vector Machines (SVM) and Random Forests (RF) have shown reasonable performance in binary text classification tasks, especially when combined with feature engineering techniques like TF-IDF and n-grams. More recently, deep learning models like Bi-LSTM and transformers (e.g., BERT) have outperformed earlier methods by capturing semantic and contextual information in texts more effectively [6][7][8].

Jiang [6] and Nakano et al. [4] have demonstrated the ability of large language models (LLMs) to detect scam-related content across platforms using contextual clues, especially in phishing and deceptive financial schemes. Their work highlights the scalability of LLMs in multilingual settings, although these models are typically trained on high-resource languages.

Kiswahili NLP and Scam Detection

Kiswahili, as a low-resource and morphologically rich language, presents unique challenges in NLP. Tokenisation, lemmatisation, and semantic embedding are more complex due to agglutination and variable word order. Efforts to create Kiswahili-specific NLP tools are relatively recent but growing. Mambina et al. [2] developed a machine-learning-based model to classify Swahili smishing attacks, using a custom-built dataset of scam messages targeting mobile money users in Tanzania. Their work showed that with well-annotated data, supervised learning algorithms can achieve promising accuracy in Swahili text classification, though they noted the limitations of

general-purpose models not trained on Swahili corpora.

Similarly, Jonathan et al. [1] implemented offensive message detection in Swahili social media content using Apache Spark, highlighting both the lack of Swahili corpora and the potential of parallel processing frameworks for large-scale language analysis.

However, none of these studies employed transformer-based models such as Swahili-BERT, which is uniquely fine-tuned for Swahili grammar and vocabulary. The current study bridges this gap by applying and comparing both classical ML and advanced NLP models for scam detection in Kiswahili.

AI-Driven Cybersecurity and Mobile Money Fraud

The integration of AI in cybersecurity applications is becoming standard practice, especially in the detection of phishing attacks, social engineering tactics, and other cyber fraud schemes. Ismail [8] and Atawneh & Aljehani [9] reviewed AI-driven cybersecurity systems and emphasised the role of real-time detection and explainability in critical environments like financial systems.

In Tanzania, Kenya, Uganda and other East African countries, mobile money services such as M-Pesa and Airtel Money dominate digital financial transactions. These platforms are increasingly targeted by scammers using SMS as the primary vector for fraud. However, region-specific cybersecurity solutions are sparse.

Abbas et al. [3] introduced a threat modelling approach to detect phishing attacks in IoT ecosystems, which, although not language-specific, outlines a scalable and proactive approach relevant to mobile financial security. Their findings support the need for domain-aware threat detection mechanisms, which this study addresses within the Tanzanian context.

Gaps in Existing Literature

Despite the evident progress in scam detection technologies, the literature reveals significant gaps in:

- **Language-specific tools for Kiswahili text classification.**
- **Annotated scam datasets that capture localised fraud patterns and dialectical variations.**
- **Deployment - focused research that examines real-time, user-centred implementation strategies for end-users in developing regions.**

This study contributes to closing these gaps by developing and evaluating a suite of ML models—including Swahili-BERT—for classifying scam messages in Kiswahili, leveraging a uniquely constructed dataset sourced from real-world Tanzanian contexts.

RESEARCH METHODOLOGY

The research methodology involved:

- **Data Collection:** Gathering real-world Kiswahili scam messages.
- **Data Annotation:** Labelling messages as genuine or fraudulent.
- **Preprocessing:** Normalisation, stop-word removal, and tokenisation.
- **Model Development:** Implementing SVM, RF, Bi-LSTM, and Swahili-BERT.
- **Evaluation:** Using accuracy, precision, recall, and F1-score.

Dataset

The dataset used in this study was designed to capture the unique linguistic and contextual features of scam communication in Kiswahili. Data was collected from multiple sources to

ensure diversity and reliability. Specifically, three main sources were utilised:

- **Publicly Available Samples** – Existing Kiswahili SMS datasets and online repositories containing examples of fraudulent and non-fraudulent mobile messages.
- **Law Enforcement Collaboration** – Anonymised scam messages obtained in partnership with the Tanzania Police Force's cybercrime unit, which provided authentic cases of mobile money fraud, phishing attempts, and other digital scams.
- **User-Contributed Data** – Data collected through structured questionnaires and community surveys, where participants shared real-life scam messages they had received.

In total, the dataset consisted of approximately 5000 messages, of which around 60% were scam-related, and 40% were legitimate. Each message was manually annotated by Swahili language experts and cybersecurity practitioners to ensure accuracy. Annotation involved labelling messages as **Scam** or **Not Scam**, and further classifying scam messages such as mobile money fraud, fake promotions, lottery scams, and phishing.

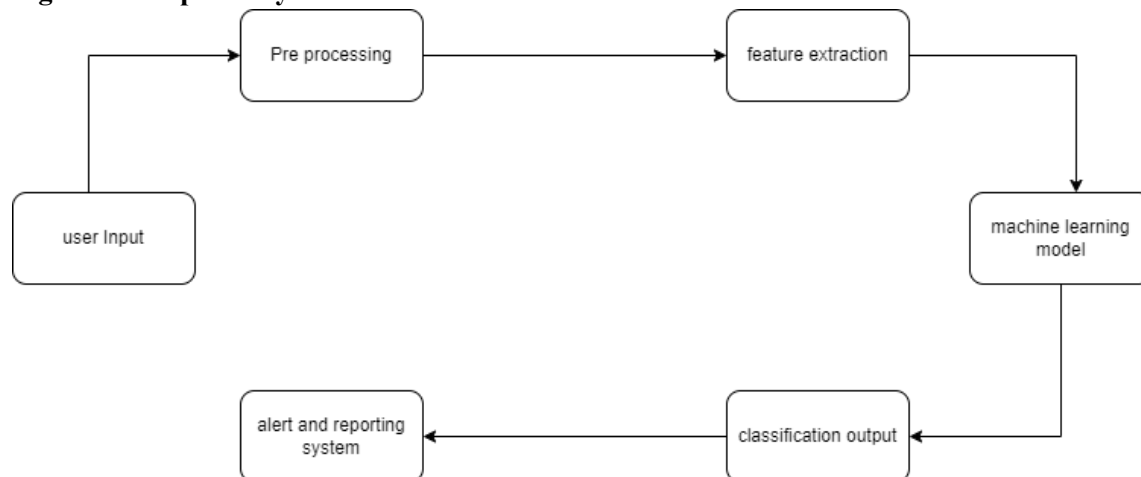
This structured dataset was then pre-processed, tokenised, and fed into the Swahili-BERT model for training and evaluation. By focusing on locally relevant messages, the dataset addressed the scarcity of Kiswahili resources in Natural Language Processing (NLP), thereby strengthening the system's applicability to Tanzania's cybersecurity landscape.

Proposed Approach

Proposed System Architecture

The proposed system architecture consists of a data preprocessing module, feature extraction engine, ML classifier, and deployment layer.

Figure 1: Proposed System Architecture



1. User Input

Input Type: SMS or Text Message

The system initiates its operation by receiving a text message sent by a user. This message serves as the primary input and may contain various content in Swahili. The message is captured by the system through a designated interface, such as an SMS gateway or a mobile application.

2. Text Preprocessing Steps:

- **Tokenisation:** The input text is segmented into individual tokens (words or terms) to enable easier analysis in subsequent stages.
- **Stopword Removal:** Commonly occurring but semantically insignificant words (e.g., "na", "ya", "kwa") are removed from the tokenised list to reduce noise and improve the focus on meaningful content.
- **Lemmatisation:** Each word is reduced to its base or dictionary form (lemma). For example, Swahili words like "kupokea" might be lemmatised to "pokea." This step standardises word forms, enhancing the accuracy of feature representation.

The preprocessing stage transforms raw, unstructured text into a cleaner and more uniform

format. This ensures that the input is suitable for feature extraction and reduces variability due to linguistic variations in the Swahili language.

3. Feature Extraction Techniques:

- **TF-IDF (Term Frequency–Inverse Document Frequency):** This statistical measure evaluates how important a word is to a message relative to a larger corpus. It helps highlight keywords that are likely to indicate scam messages.
- **Word Embeddings (e.g., Word2Vec, GloVe):** Words are represented as dense vectors in a continuous vector space, capturing semantic relationships between words. This allows the system to understand context and similarity between terms even if they are not identical.

At this stage, the preprocessed text is transformed into a structured numerical format that can be understood by the machine learning model. This representation preserves the semantic and statistical features of the message content, enabling more accurate classification.

Proposed Machine Learning Model

- **Input:** Numerical feature vectors obtained from the Feature Extraction step

- **Model Type:** Supervised learning model trained on a labelled dataset of Swahili scam and non-scam messages

The core of the system is a machine learning classifier that analyses the extracted features. It has been trained on historical SMS data, where each message is labelled as either “Scam” or “Non-Scam.” Using this knowledge, the model predicts the label of incoming messages based on patterns it has learned.

Classification Possible Output

i). Scam

ii). Non-Scam

The machine learning model outputs a classification label that indicates whether the incoming message is potentially fraudulent ("Scam") or safe ("Non-Scam"). This result forms the basis for subsequent action or alert.

Figure 2: Use Notification

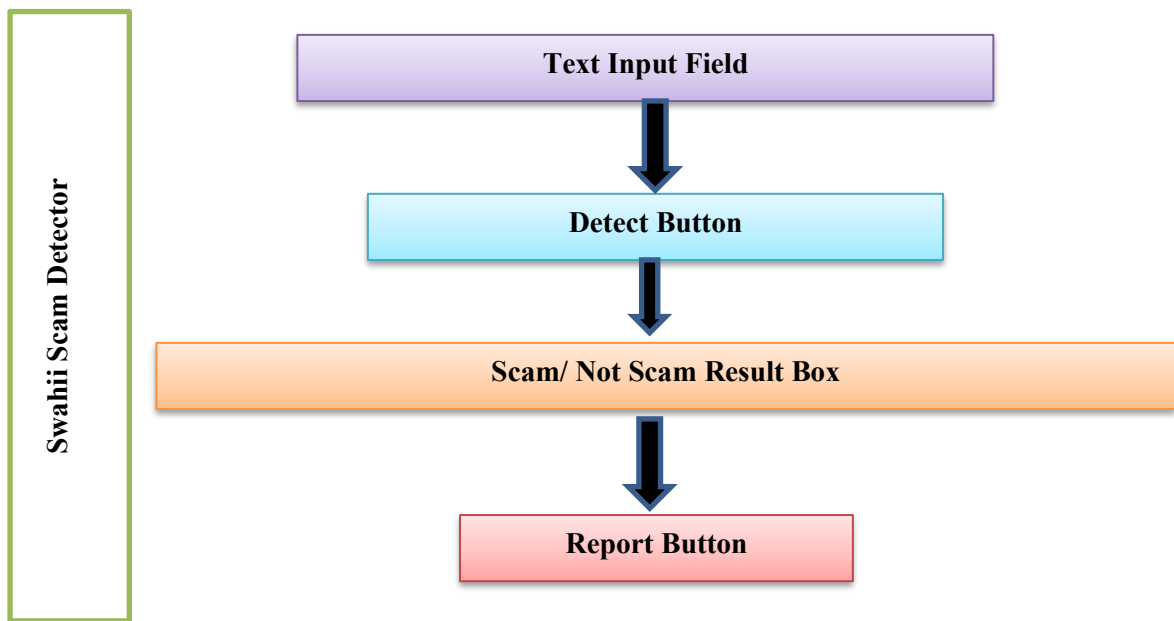


- **User Notification:** If a message is classified as a scam, the system may display a warning notification to the user, cautioning them against interacting with the content.
- **Message Blocking:** In cases where automatic protection is enabled, the system can prevent the message from reaching the

user's inbox or take other security measures, such as reporting the sender.

This optional module enhances user safety by taking proactive steps when a scam is detected. It adds a protective layer to the classification system, helping mitigate risks associated with SMS fraud and social engineering attacks.

Figure 3: Proposed Model



Backend System Code (Flask API)

A Flask-based API was implemented to enable integration with SMS platforms and mobile apps. It handles incoming messages, preprocesses text, applies the trained ML model, and returns classification results.

Model Performance

The developed models were evaluated using standard performance metrics, including Accuracy, Precision, Recall, and F1-Score. The Table below summarises the performance of each model.

RESULT AND DISCUSSION

Table 1: Performance of Machine Learning Models on Kiswahili Scam Detection

Models	Accuracy	Precision	Recall	F1-Score
SVM	91.1%	92.0%	90.1%	91.0%
Random Forest	93.2%	93.6%	92.8%	93.2%
Bi-LSTM	94.2%	94.5%	94.1%	94.3%
Swahili-BERT	96.3%	96.5%	96.0%	96.2%

Summary of Results:

- Swahili-BERT achieved the **highest detection accuracy** (96.3%) compared to all other models.
- Deep learning models (Bi-LSTM and Swahili-BERT) consistently outperformed traditional ML models (SVM and Random Forest).
- Scam messages were found to commonly exhibit features such as **urgency, requests for personal information, and poor grammar**.

Compliance: Adhering to Tanzanian cybersecurity regulations.

DISCUSSION

The findings of this study clearly demonstrate the viability and effectiveness of applying machine learning (ML) and Natural Language Processing (NLP) techniques to the problem of scam message detection in Kiswahili, a low-resource language. By comparing traditional ML models (SVM and Random Forest) with deep learning architectures (Bi-LSTM and Swahili-BERT), the results show a progressive improvement in classification performance, with Swahili-BERT achieving the

highest accuracy (96.3%), precision (96.5%), recall (96.0%), and F1-score (96.2%).

Performance of Machine Learning Models

The performance of the Swahili-BERT model highlights the power of transformer-based architectures, particularly in handling linguistic complexities such as agglutination, variable word order, and rich morphology found in Kiswahili. Unlike traditional models, which rely heavily on hand-crafted features and may struggle to capture contextual meaning, Swahili-BERT benefits from pretraining on a language-specific corpus and fine-tuning with task-specific data. The model's superior performance reinforces the importance of using language-tailored resources for NLP tasks in low-resource languages.

In contrast, traditional models like SVM and Random Forest still showed respectable performance, with accuracies above 90%. These models are computationally lightweight and may remain useful in resource-constrained environments. However, their lower performance, especially in capturing context and semantic nuances, limits their scalability and robustness in real-world deployments.

Insights from the Dataset

The dataset, compiled from public sources, law enforcement, and user submissions, revealed key features common to scam messages in Kiswahili. These include urgency cues (e.g., "haraka sana"), financial incentives (e.g., "umeshinda milioni"), personal information requests, and poor grammatical structure. Such insights not only informed model training but also provided valuable linguistic patterns for future rule-based systems or hybrid approaches.

Importantly, the collaboration with the Tanzania Police Force ensured that the dataset reflected actual scam strategies used by cybercriminals. This adds both practical relevance and authenticity to the model, enhancing its potential for real-world effectiveness.

Practical Implications

The deployment of this system through a Flask API demonstrates the model's readiness for integration into mobile applications and SMS gateways. This architecture can support telecom companies, financial institutions, and law enforcement agencies in identifying and mitigating mobile money fraud in real-time.

Moreover, the system's user-notification and optional message-blocking features enhance its practical utility, providing users with immediate warnings while offering backend support for automated scam filtering.

This research also aligns with the Tanzanian government's efforts to improve digital safety, especially in the financial sector, where mobile money plays a critical role. The integration of such technologies could significantly reduce fraud cases, enhance trust in mobile platforms, and contribute to national cybersecurity objectives.

Challenges and Limitations

Despite the promising results, the study encountered several challenges:

Data scarcity: Kiswahili remains a low-resource language, with limited annotated corpora available. The construction of a high-quality dataset required significant manual effort, particularly in annotation and classification.

Dialectical and informal variations: Kiswahili messages often contain regional dialects or informal spellings, which may reduce model performance. Future systems must account for these variations through more diverse training data.

Evolving scam patterns: Cybercriminal tactics are constantly evolving. Models trained on historical data may become outdated unless continually updated. Incorporating mechanisms for ongoing data collection and retraining is critical for long-term effectiveness.

Ethical and Privacy Considerations

The deployment of such a system must be accompanied by strong data protection and privacy safeguards. As the system processes sensitive user messages, it must comply with data privacy regulations and ethical guidelines, including anonymisation, user consent, and transparency in message classification.

Additionally, while the system can flag potential scams, it should not replace human oversight in legal or enforcement decisions. Misclassifications, although rare, could have consequences if users wrongly ignore legitimate messages or law enforcement takes action on false positives.

CONCLUSION

This study has demonstrated that machine learning and Natural Language Processing (NLP) techniques can be effectively applied to detect scam messages written in Kiswahili—a low-resource and morphologically rich language. By developing and evaluating multiple models, including traditional classifiers (SVM and Random Forest), deep learning (Bi-LSTM), and a transformer-based model (Swahili-BERT), we found that Swahili-BERT significantly outperformed the others, achieving an accuracy of 96.3%. Its superior performance highlights the importance of using language-specific transformer models for improved text classification in underrepresented languages.

The creation of a Kiswahili scam message dataset, sourced through a combination of public data, user contributions, and law enforcement collaboration, addressed a major gap in available resources for Kiswahili NLP. The proposed system architecture, supported by a Flask API, enables real-time integration with mobile and SMS platforms, offering a scalable and practical solution to combat mobile money fraud in Tanzania.

Beyond its technical contributions, this research advances the broader goals of digital security, user protection, and technological empowerment in

East Africa. It demonstrates that with the right tools and localised data, AI-driven solutions can be successfully developed for low-resource settings, providing both societal and policy impact.

However, continued efforts are needed to expand annotated datasets, improve NLP tools for Kiswahili, and ensure ethical deployment. Real-time systems must be continuously updated to adapt to evolving scam tactics and maintain relevance in real-world applications.

Future Recommendations

Based on the findings and limitations of this study, several recommendations are proposed to enhance the development, deployment, and sustainability of Kiswahili-based scam detection systems in the future:

1. Expansion of Annotated Datasets

To improve model generalisation and accuracy, it is essential to expand the current dataset with a larger and more diverse set of Kiswahili messages. Future efforts should include regional dialects, informal language variations, and emerging scam techniques to ensure broader language and threat coverage.

2. Development of Kiswahili NLP Tools

There is a pressing need for the development of additional Kiswahili NLP tools such as lemmatisers, part-of-speech taggers, named entity recognition (NER) systems, and dependency parsers. These tools would enable more sophisticated language understanding and enhance performance in downstream tasks like fraud detection, sentiment analysis, and content moderation.

3. Real-time System Deployment

Stakeholders—including telecom operators, financial service providers, and government agencies—should prioritise the deployment of the proposed system in real-time environments. Integration with SMS gateways and mobile apps can help filter scam messages before they reach

users, improving immediate protection and response.

4. Periodic Model Updating

Since scam tactics evolve rapidly, it is critical to implement continuous learning strategies. Periodic retraining of models using newly collected scam messages will help maintain detection accuracy and system relevance over time.

5. Cross-Language and Multilingual Model Development

Given the multilingual nature of digital communication in East Africa, future models should explore multilingual or cross-lingual capabilities, enabling the detection of scam messages in mixed-language formats (e.g., Kiswahili-English code-switching).

6. User Awareness and Education Programs

Technological solutions should be complemented by public awareness campaigns to educate users on recognising and reporting scam messages. Collaboration with the Tanzania Police Force and telecom providers can enhance community engagement and improve early scam detection through user reports.

7. Policy and Regulatory Support

The government and relevant regulatory bodies should establish supportive frameworks for the ethical deployment of AI-based detection systems. This includes guidance on data privacy, transparency, accountability, and secure data sharing across institutions [12].

8. Integration with Law Enforcement Systems

Future work should focus on integrating scam detection platforms directly with law enforcement databases and reporting tools [12]. This would facilitate faster investigation, case management, and legal action against scammers.

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